

Perfect algebraic geometry

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p fixed prime

$$/\mathbb{F}_p \operatorname{Spec} A[\frac{1}{p}] \xrightarrow{\phi} \operatorname{Spec} A[\frac{1}{p}]$$

$$\text{Given } A \xrightarrow{\phi} A \\ x \mapsto x^p$$

$$\operatorname{Spec} A \xrightarrow{\phi} \operatorname{Spec} A$$

identity on topological space

Universal homeomorphism

$$X \xrightarrow{\phi} X \\ \downarrow \mathbb{F}_p$$

affine (integral) universal homeomorphism

$$\underline{\text{Def}} \quad X^{\text{perf}} = X^{p^{-\infty}} := \varprojlim (X \xleftarrow{\phi} X \xleftarrow{\phi} \dots)$$

$\downarrow$ affine integral universal homeomorphism
 X

$$(\operatorname{Spec} A)^{\text{perf}} = \operatorname{Spec} \left(\varinjlim A \xrightarrow{\phi} A \xrightarrow{\phi} \dots \right)$$

$$\underline{\text{Ex.}} \quad (\mathbb{A}_{\mathbb{F}_p}^1)^{\text{perf}} = \operatorname{Spec} \mathbb{F}_p[X, X^{1/p}, X^{1/p^2}, \dots]$$

not noetherian! $(X, X^{1/p}, \dots)$ is not finitely generated

Rank X qcs

$$\operatorname{Vect}(X^{\text{perf}}) = \varinjlim \left(\operatorname{Vect}(X) \xrightarrow{\phi^*} \operatorname{Vect}(X) \rightarrow \dots \right)$$

§ perfectness & universal homeomorphism

Thm 0 $f: X \rightarrow Y$ map of (arbitrary) schemes.

f univ. homeo. $\Leftrightarrow f$ integral, surjective, radicial.

every geom. fiber has 1 pt
w/ same res. field

Thm 1 For schemes $/\mathbb{F}_p$,

f univ. homeo. $\Leftrightarrow f^{\text{perf}}: X^{\text{perf}} \rightarrow Y^{\text{perf}}$ being an isom.

Thm 2 If $A \subsetneq B$ is (not nec. $/\mathbb{F}_p$) ring extension, and $\text{Spec } B \rightarrow \text{Spec } A$

univ. homeomorphism. $\Rightarrow \exists b \in B - A$ s.t. either ① $b^2, b^3 \in A$ or

② $b^p, pb \in A$ for some p

Pf of Thm 2

① Reduce to when B is finite over A .

$A \subsetneq B' \subset B$ s.t. B' gen. by single elt.

$\underbrace{A \subsetneq B'}_{\text{finite}} \xrightarrow{\text{univ. homeo.}} \text{Spec } B' \rightarrow \text{Spec } A$

lying over $\Rightarrow$ both univ. homeom.

$I = \text{Ann}_A(B/A) \Rightarrow V(I) = \text{Supp}_A(B/A)$ $\xrightarrow{\text{B local ring}}$

② Reduce to the case when A local ring, $\mathfrak{m}_A \subset A$,

$V(I) = \text{Supp}_A(B/A) = \{\mathfrak{m}_A\}$.

Take $\mathfrak{p} \in V(I)$ a minimal prime. Take $A_{\mathfrak{p}} \subset B_{\mathfrak{p}}$.

$$\Rightarrow \text{Supp}_{A_{\mathfrak{p}}}(B_{\mathfrak{p}}/A_{\mathfrak{p}}) = \{\mathfrak{p} \cdot A_{\mathfrak{p}}\}$$

Once $b_0 \in B_{\mathfrak{p}} - A_{\mathfrak{p}}$ s.t. $\begin{cases} \textcircled{1} \\ \text{or} \\ \textcircled{2} \end{cases} \Rightarrow b_0 = \frac{b}{s} \text{ for } s \in A - \mathfrak{p}, b \notin A.$

$$b^2 \cdot (\text{sth. in } A - \mathfrak{p}) \in A$$

$$b^3 \cdot (\text{---}) \in A$$

$$\textcircled{3} \quad IB \subset A \cap m_B = m_A \subset m_B \quad I = \text{Ann}_A(B/A), \quad \sqrt{I} = m_A, \quad \sqrt{I \cdot B} = m_B$$

Case 1: $m_B = m_A$.

$A \neq B \Rightarrow A/m_A \subsetneq B/m_B$ purely inseparable

$\rightarrow$ in char p for some p , $\exists \bar{b} \in B/m_B$ s.t. $\bar{b}^p \in A/m_A$

Take b lift of $\bar{b}$, $b^p \in A$, $pb \in m_B \subset A$.

Case 2: $m_B \neq m_A$.

Take $b_0 \in m_B \setminus m_A = m_B \setminus A$.

$$b_0^n \in I \cdot B \subset A \text{ for } n \gg 0$$

$$b_0^{n+1} \in A$$

$b_0^m \notin A$ for m maximal, then $b = b_0^m$ satisfies $b^2, b^3 \in A$, $b \notin A$.

Pt of Thm 1. $\Leftrightarrow$ f univ. homeo $\Rightarrow$ affine $\rightarrow$ reduce to rings

$A \hookrightarrow B$ both perf, inj. because $\text{Spec } B \rightarrow \text{Spec } A$, A reduced
if not isom $\rightarrow \exists b \in A$ s.t. either $\textcircled{1} b^2, b^3 \in A$
or $\textcircled{2} pb, b^p \in A$
 $\Rightarrow b^p \in A \Rightarrow b \in A$.

$$\Leftarrow \begin{array}{ccc} X^{\text{perf}} & \xrightarrow{\text{u.h.}} & Y^{\text{perf}} \\ \text{u.h.} \downarrow & & \downarrow \text{u.h.} \\ X & \longrightarrow & Y \end{array}$$



§ Base change

Thm 3. If $B \leftarrow A \rightarrow C$ are perfect rings, then $\text{Tor}_i^A(B, C) = 0$ for $i > 0$

$$\text{i.e. } B \underset{A}{\otimes} C = B \underset{A}{\otimes} C$$

Cor. If $f: Y \rightarrow X$ qqs of perfect schemes, F/Y coh.

$$\begin{array}{ccc} Y' & \xrightarrow{g'} & Y \\ f' \downarrow & & \downarrow f \\ X' & \xrightarrow{g} & X \end{array} \quad \begin{array}{l} X' \text{ perfect} \Rightarrow Y' \text{ perfect} \\ \Rightarrow Lg^* Rf_* F \xrightarrow{\sim} Rf'_* Lg'^* F \end{array}$$

Pt of Thm 3. Both sides commute w filtered colim in B

$\leadsto$ reduce to B is "perfectly finitely presented" over A

$$B \simeq (B_0)^{\text{perf}} \text{ for } A \xrightarrow{\text{pt.}} B_0$$

$$\leadsto B = A[x^{1/p^\infty}] \leadsto B \text{ is free } / A \quad \square$$

$$\leadsto B = A/(f, f^{1/p}, \dots)$$

$$\text{Claim } I = (f, f^{1/p}, \dots)$$

$$\lim_{\longrightarrow} \left(A \xrightarrow{f^{1-\frac{1}{p}}} A \xrightarrow{f^{\frac{1}{p}-\frac{1}{p^2}}} A \xrightarrow{f^{\frac{1}{p^2}-\frac{1}{p^3}}} \dots \right) \text{ is an isom.}$$

Pt Surjectivity is clear. Injectivity: $x \in A$, $f^{1/p^n} x = 0 \Rightarrow f^{\frac{1}{p^n}} x^p = 0$
 $\Rightarrow f^{\frac{1}{p^{n+1}}} x = 0 \quad \square$

$$I = \varinjlim \left(A \xrightarrow{f^{1-\frac{1}{p}}} A \xrightarrow{f^{\frac{1}{p}-\frac{1}{p^2}}} A \rightarrow \dots \right)$$

$$I \overset{\mathbb{L}}{\otimes}_A C = \varinjlim \left(C \xrightarrow{f^{1-\frac{1}{p}}} C \xrightarrow{f^{\frac{1}{p}-\frac{1}{p^2}}} C \rightarrow \dots \right) = IC \left((f, f^{1/p}, \dots) \text{ in } C \right)$$

$$\rightarrow B_A \overset{\mathbb{L}}{\otimes} C = C/IC = \text{cofib} \left(I \overset{\mathbb{L}}{\otimes}_A C \rightarrow C \right) = B_A \overset{\mathbb{L}}{\otimes} C.$$

Thm 4. Let $f: Y \rightarrow X$ be a proper surjective map of noetherian $\mathbb{F}_p$ -schemes

$Z \subset X$ closed, $U = X - Z$ s.t. $f^{-1}(U) \xrightarrow{\sim} U$ is an isom. $E = f^{-1}(Z)$.

$$E \subset Y$$

$$\begin{array}{ccc} f \downarrow & & f \downarrow \\ Z & \subset & X \end{array}$$

$$\boxed{1} \Rightarrow \mathcal{O}_X^{\text{perf}} \rightarrow \mathcal{O}_Z^{\text{perf}} \oplus Rf_* \mathcal{O}_Y^{\text{perf}} \rightarrow Rf_* \mathcal{O}_E^{\text{perf}}$$

is a distinguished triangle.

$\boxed{2}$ There is an equiv. of cat. $\text{Vect}(X)^{\text{rat}} \xrightarrow{\sim} \text{Vect}(Y)^{\text{rat}} \times_{\text{Vect}(E)^{\text{rat}}} \text{Vect}(Z)^{\text{rat}}$ (De Bois ?)

Remark. Both statements can be upgraded to v -cover.

Pf of Thm 4 I. Can assume X is affine.

$$\textcircled{1} \left[\begin{array}{l} 0 \rightarrow \mathcal{O}(X)^{\text{perf}} \rightarrow \mathcal{O}(Z)^{\text{perf}} \oplus \mathcal{O}(Y)^{\text{perf}} \rightarrow \mathcal{O}(E)^{\text{perf}} \rightarrow 0 \\ R^i f_* \mathcal{O}_Y^{\text{perf}} \simeq R^i f_* \mathcal{O}_E^{\text{perf}}, \quad i > 0 \end{array} \right]$$

$$\textcircled{2} \begin{array}{c} z' \rightarrow Z \\ \uparrow \text{finite} \\ Y \xrightarrow{g} X' \rightarrow X \end{array} \quad \text{Stein factorisation}$$

$$\mathcal{O}(Y) = \mathcal{O}(X') \text{ by def'n.}$$

$$\mathcal{O}(E)^{\text{perf}} = \mathcal{O}(Z')^{\text{perf}} \text{ by base change.}$$

$\Rightarrow$ reduce to the case when f is finite.

Everything is affine.

WTS $\mathcal{O}(x) \rightarrow \ker \left(\mathcal{O}(Z) \oplus \overset{A}{\underset{ii}{\mathcal{O}(Y)}} \rightarrow \mathcal{O}(E) \right)$ is an isom. upon perfection.

$\Leftrightarrow \mathcal{O}(x) \rightarrow A$ a univ. homeom.

$\int \mathcal{O}(x) \rightarrow A$ is finite ($\mathcal{O}(x)$ noetherian)

fibres of $\text{Spec } A \rightarrow \underset{\text{Spec } \mathcal{O}(x)}{X}$ being univ. homeom.

Given $\text{Spec } k \rightarrow \text{Spec } \mathcal{O}(x)$

$$A \otimes_{\mathcal{O}(x)} k \xrightarrow{\text{base change thm}} \ker \left((\mathcal{O}(Z) \oplus \mathcal{O}(Y)) \otimes_{\mathcal{O}(x)} k \rightarrow \mathcal{O}(E) \otimes_{\mathcal{O}(x)} k \right) \xrightarrow{\text{divide into 2 cases:}} k \text{ upon perfection}$$

divide into 2 cases:

$\text{Spec } k$ lands in U or Z .